

Customer Churn Risk

Short Report

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For LearnDepth ML Internship

Objective: Predict whether a subscription customer will churn in the next billing cycle.

- ``target = 1`` → positive / risk / churn event
- ``target = 0`` → negative / non-event (customer stays)

I. Problem Statement

The business problem is to predict using machine learning whether a subscription customer will churn or cancel their subscription in the next billing cycle.

II. Data

The dataset includes 1000 rows and 7 columns. It is a *binary classification* with 6 features and 1 target.

– **Binary classification** is a machine learning task that sorts data into one of two distinct categories or classes.

Feature	Description	Type	Range
tenure_months	Time for subscription	Numeric	1-60 months
monthly_charges	Monthly subscription fee	Numeric	16.81-1530.68
support_tickets	Number of support tickets opened	Numeric	0-11
avg_session_minutes	Average session duration per visit	Numeric	2-103.35 minutes
late_payments	Amount of late payments	Numeric	0-10
contract_months	Contract length (in months)	Numeric	1-60 months
target	Churn status	Binary	0 = no churn (continue) 1 = churn

```
df['target'].value_counts() # checks target distribution
```

It is revealed that the dataset is perfectly balanced without class imbalance issues – both have 50%.

III. Preprocessing

Data Quality Check

- Missing values and duplicate rows = 0 – imputation or deduplication aren't necessary
 - Data types = all numeric – encoding aren't necessary
 - Class balance = balanced
 - Outliers = found in monthly_charges
- **Outliers** are data points that significantly differ from the rest of the dataset.

Preprocessing

1. Split features and target
2. Train/Test Split (80/20 with stratification)
3. Feature scaling/standardization

IV. Model

Baseline Model serves as a minimum performance benchmark that the actual model must beat.

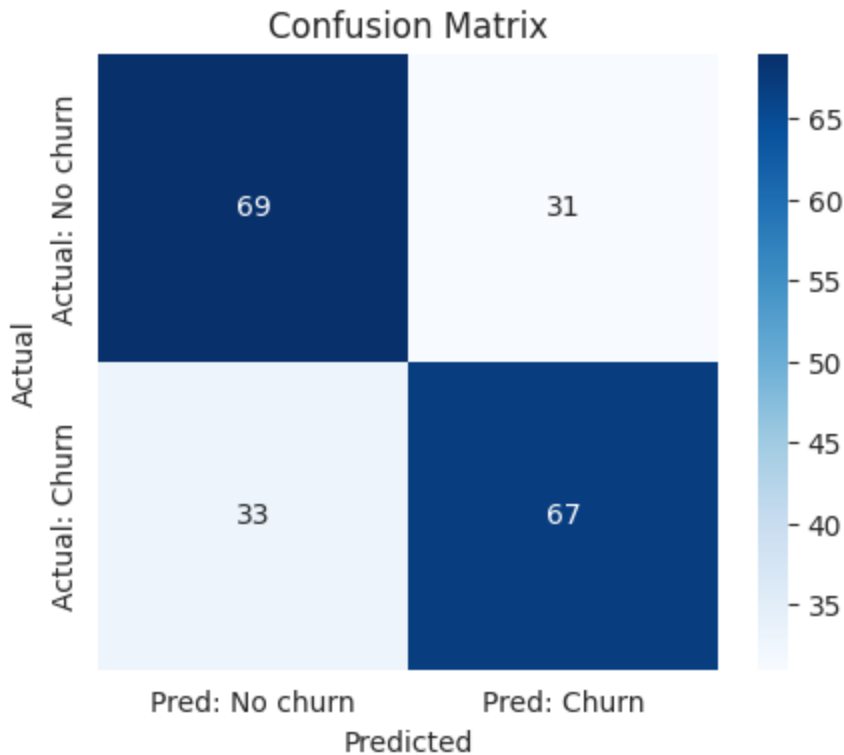
Logistic Regression Model serves as a statistical method for binary classification – predicting which of two categories an observation belongs to.

V. Results

Performance Metrics

1. Accuracy - 68.00% - Model correct 68% of predictions (vs 50% baseline)
2. Precision - 68.37% - When predicting churn, 68% are **actually** churn
3. Recall - 67.00% - Model catches 67% of actual churners
4. F1-Score - 67.88% - Harmonic mean of precision and recall
5. ROC-AUC - 74.95% - Model distinguishes churn vs not-churn well

Confusion Matrix



1. True Negatives (TN): 69 – correctly predicted STAY
2. False Positives (FP): 31 – predicted churn, but actually STAYED
3. False Negatives (FN): 33 – predicted stay, but actually CHURNED
4. True Positives (TP): 67 – correctly predicted CHURN

Feature Coefficients

	feature	coefficient	odds_ratio
1	monthly_charges	0.646803	1.909427
5	contract_months	0.536243	1.709573
3	avg_session_minutes	0.508222	1.662333
2	support_tickets	-0.447767	0.639053
4	late_payments	-0.428239	0.651655
0	tenure_months	-0.351499	0.703632

Monthly_charges -> increases risk on churn

Contract_months -> increases risk of churn

Avg_session_minutes -> increases risk of churn

Support_tickets -> decreases risk of churn

Late_payments -> decreases risk on churn

Tenure_months -> decreases risk on churn

INTERPRETATION – The Logistic Regression model achieved 68% accuracy and 74.95% ROC-AUC, significantly outperforming the baseline dummy classifier (50%)

accuracy). The model successfully identifies key churn drivers, with monthly charges being the strongest predictor (Odds Ratio = 1.91), followed by contract length (OR = 1.71) and average session duration (OR = 1.66). Interestingly, support tickets (OR = 0.64), late payments (OR = 0.65), and tenure (OR = 0.70) show negative coefficients, meaning they are associated with lower churn risk. While some findings are intuitive (high charges and short tenure increase churn), others are counter-intuitive (longer contracts and high engagement increase churn), likely due to the synthetic nature of the dataset. The model demonstrates practical value for customer retention strategies but requires validation on real-world data before operational deployment.

VI. Limitations

Data

1. As data is synthetic and generated by computer (not real customers), this can affect how it reflects actual reality
2. As there are only 6 predictors, other churn factors aren't captured
3. There are outliers that are unrealistically wide in monthly charges

Model

1. Linear assumption cannot capture non-linear relationships/interactions
2. Counter-intuitive signs in support_tickets and late_payments that shows unexpected results (negative coefficients)
3. Coefficients show association, NOT causation
4. No feature engineering as there are no polynomial terms or interaction features that are created

Generalization Risk

1. Customer behavior can change over time
2. Model may not generalize to different customer segments
3. Temporal dependency as there are no time component in features

VII. Conclusion

The Logistic Regression model successfully predicts customer churn with **68% accuracy and a 74.95% ROC-AUC score**, significantly outperforming the baseline model (50% accuracy).

Monthly charges emerged as the strongest churn predictor, with **higher fees** increasing churn risk by 91%, while **customer tenure** serves as the strongest protective factor, with **longer-tenured customers** being 30% less likely to churn.

Surprisingly, support tickets and late payments showed **negative coefficients**, meaning customers who contact support or have late payments are actually less

likely to churn—likely reflecting the synthetic nature of the dataset rather than real-world behavior.

Key counter-intuitive findings such as longer contracts and higher session activity being associated with increased churn risk highlight the need for validation on real customer data before operational deployment.

Overall, the model does provide valuable insights for customer retention strategies . But it should be treated as a proof of concept and screening tool, not as a definitive decision-making system, until validated with actual business data that comes from real life.